

Non-Equal-Weight Risk Allocation

Cross-Scheme Comparison

MetaFVG – Spearman Mean-Reversion – 14-Instrument Diversified Portfolio

Fixed 2% Notional Sizing, 5-Year Sample

Motivation

The 14-instrument book (7 FX Majors + US500/GER40/JP225/HK50 + USDSGD/EURNOK/USDZAR) was assembled to be fundamentally diversified and is currently blended with equal weights (1/14 each), realizing an average pairwise correlation of **0.0156** and a Sharpe of **0.518**. This report tests whether replacing equal weighting with a correlation-aware allocation scheme can push realized correlation down further without giving up risk-adjusted return. Four schemes were tested, each with its own detailed report in this directory: [inverse_vol.pdf](#), [erc.pdf](#), [hrp.pdf](#), [inverse_avg_corr.pdf](#).

Headline Result

None of the four schemes beat equal weighting on Sharpe. Only one (Inverse Average-Correlation) reduced realized correlation *and* improved the diversification ratio, and even it gave up some Sharpe. The other three schemes made both correlation *and* risk-adjusted return worse.

Ranked by Sharpe (descending). Weighted avg. correlation and diversification ratio (DR) are computed under each scheme's own weight vector.

Scheme	Wtd. Avg. Corr.	DR	Sharpe	Sortino	Calmar	CAGR	Max DD
Equal Weight (baseline)	0.0156	2.081	0.518	0.817	0.343	0.0183%	-0.0533%
Inverse Avg-Correlation	0.0118	2.110	0.483	0.760	0.315	0.0168%	-0.0533%
Inverse-Volatility	0.0159	1.809	0.321	0.504	0.200	0.0107%	-0.0533%
Equal Risk Contribution	0.0142	1.831	0.311	0.488	0.191	0.0102%	-0.0533%
Hierarchical Risk Parity	0.0170	1.538	0.170	0.265	0.106	0.0057%	-0.0533%

Why every Max Drawdown row is identical. This was checked and confirmed, not a bug: on any day where only one instrument in the book has a closed trade, the per-day weight-renormalized blend formula ($(\sum_i w_i r_{i,t})/(\sum_i w_i)$) collapses to exactly that instrument's raw return, independent of the weight vector. This 14-instrument book has sparse, largely non-overlapping trade timing, and the single worst day driving the -0.0533% max drawdown happens to be one of these single-instrument days – so it is identical across every weighting scheme by construction. Differences between schemes only show up on multi-instrument-active days, which is exactly where Sharpe/Sortino/Calmar diverge.

Symbol	Equal	Inv-Vol	ERC	HRP	Inv-Avg-Corr
AUDUSD	7.14	6.54	5.94	4.82	5.28
EURUSD	7.14	10.50	10.28	12.43	8.11
GBPUSD	7.14	8.90	7.95	8.70	4.50
NZDUSD	7.14	5.71	5.41	4.48	6.13
USDCAD	7.14	11.20	10.94	14.48	8.11
USDCHF	7.14	8.34	9.03	7.84	8.11
USDJPY	7.14	7.09	7.40	7.24	8.11
US500	7.14	4.31	4.01	2.25	5.88
GER40	7.14	4.50	4.10	2.22	5.19
JP225	7.14	2.92	2.89	1.03	8.11
HK50	7.14	3.11	3.37	1.39	8.11
USDSGD	7.14	13.51	13.62	20.60	8.11
EURNOK	7.14	8.39	9.94	9.69	8.11
USDZAR	7.14	4.95	5.11	2.83	8.11

Weight Allocation by Scheme (%)

Per-Scheme Findings

Inverse-Volatility (naive risk parity)

Concentrates capital into the lowest-standalone-vol FX legs (USDSGD 13.5%, USDCAD 11.2%, EURUSD 10.5%) at the expense of the equity indices (JP225 2.9%, HK50 3.1%). Weighted correlation barely moves ($0.0156 \rightarrow 0.0159$, marginally *worse*) because the scheme ignores cross-instrument correlation entirely – it only equalizes standalone volatility, which is a different axis from diversification. Sharpe falls 38% ($0.518 \rightarrow 0.321$).

Equal Risk Contribution (true risk parity)

Uses the full covariance matrix and converges cleanly (11 iterations, risk-contribution spread $\sim 5 \times 10^{-8}$ relative to target). Weighted correlation does fall ($0.0156 \rightarrow 0.0142$, $\sim 9\%$ relative reduction) but the diversification ratio gets worse ($2.081 \rightarrow 1.831$): ERC deliberately defunds the highest-standalone-variance legs (equity indices) to equalize risk contribution, and those are exactly the legs providing this book’s real diversification benefit. Sharpe falls 40% ($0.518 \rightarrow 0.311$).

Hierarchical Risk Parity

The cluster structure is fundamentally sensible – the dendrogram groups JP225/US500/GER40 (equity indices) and AUDUSD/NZDUSD (commodity bloc) as expected. But the recursive bisection then allocates *within* each cluster by inverse standalone variance, which again defunds the equity-index legs (JP225 down to 1.0%, HK50 to 1.4%, versus 7.14% equal) – the instruments providing the most real cross-asset diversification end up most starved of capital. This is the worst scheme tested on every metric: correlation gets *worse* ($0.0156 \rightarrow 0.0170$), diversification ratio drops most ($2.081 \rightarrow 1.538$), and Sharpe collapses by two-thirds ($0.518 \rightarrow 0.170$).

Inverse Average-Correlation

The only scheme that improved both target metrics: weighted correlation fell to 0.0118 (the lowest of all four) and the diversification ratio rose to 2.110 (the only scheme to beat the equal-weight baseline on this axis). The correlation floor (clipped at 0.02 before inverting) bound for 9 of 14 instruments,

collapsing them to a common weight – this pulled capital away from the book’s best individual performers (GBPUSD, AUDUSD, NZDUSD, standalone Sharpe ~ 0.66 – 0.76) toward several historically negative-Sharpe legs that happened to have low average correlation. Net effect: Sharpe fell only 7% ($0.518 \rightarrow 0.483$), by far the smallest degradation of the four schemes, for a real reduction in realized correlation.

Conclusion

Equal weighting remains the best choice for live use – it has the highest Sharpe of anything tested, and the portfolio’s low correlation (0.0156) was already earned upstream, by the fundamental-driver instrument selection, not by clever weighting. Correlation-aware weighting schemes here consistently trade away return for a diversification statistic that was already good. If a lower-correlation allocation is wanted for other reasons (e.g. tail-risk budgeting across strategies, not just this one), **Inverse Average-Correlation** is the only scheme that earns its keep: real correlation and diversification-ratio improvement for the smallest Sharpe cost, and it is the most promising candidate for further tuning (e.g. a softer floor than 0.02, or blending its weights with equal weight rather than using them outright).